Continuous Time Fourier Transform (CTFT)

$$X(\omega) \triangleq \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\, d\omega$$

Dirichlet Condit^n: (Sufficient but not necessary)

1) $\int_{-\infty}^{\infty} |x(t)|\, dt < \infty$

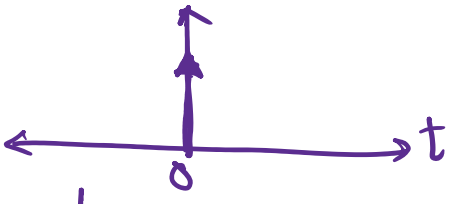
2) $x(t)$ has finite no. of maxima & minima over any finite interval

3) " " of finite discontinuities " "

$\Rightarrow$ $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\, d\omega$, except at the pts. of discont.

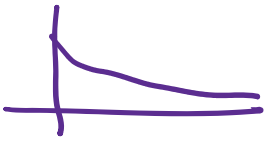
where value = avg. of values on either sides " " .

Ex 1: $x(t) = \delta(t)$



$$\Rightarrow X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t}\, dt = 1$$

$$\therefore \quad \delta(t) \longleftrightarrow 1$$



2) $e^{-at} u(t) = x(t)$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t}\, dt = \int_{0}^{\infty} e^{-at} e^{-j\omega t}\, dt = \int_{0}^{\infty} e^{-(a+j\omega)t}\, dt$$

$$= \frac{e^{-(a+j\omega)\infty} - 1}{-(a+j\omega)} = \frac{1 - e^{-(a+j\omega)\infty}}{a+j\omega} = \frac{1}{a+j\omega}, \quad Re\{a\} > 0$$

$$\therefore\ e^{-at}u(t) \longleftrightarrow \frac{1}{a+j\omega}\ ,\ Re\{a\}>0$$

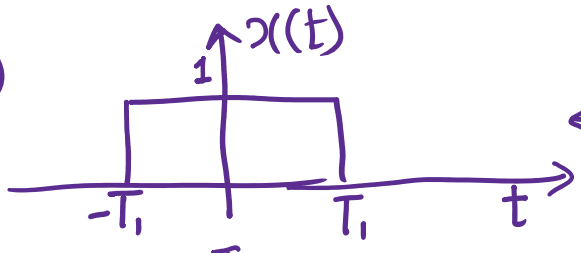
3) $e^{-a|t|} = x(t)$

$$X(\omega) = \int_{-\infty}^{0} e^{-a(-t)} e^{-j\omega t}\,dt + \int_{0}^{\infty} e^{-at} e^{-j\omega t}\,dt$$

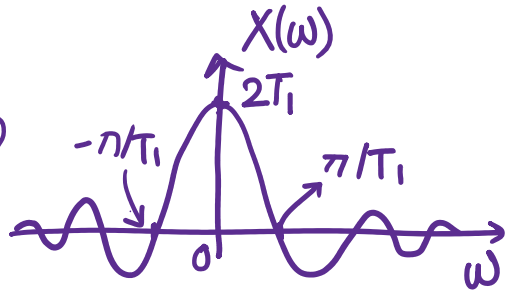
$$= \underbrace{\int_{-\infty}^{0} e^{at} e^{-j\omega t}\,dt}_{\frac{1}{a-j\omega},\ Re\{a\}>0} + \underbrace{\int_{0}^{\infty} e^{-at} e^{-j\omega t}\,dt}_{\frac{1}{a+j\omega},\ Re\{a\}>0}$$

$$\Rightarrow X(\omega) = \frac{1}{a-j\omega} + \frac{1}{a+j\omega} = \frac{2a}{a^2+\omega^2}\ ,\ Re\{a\}>0$$

4)



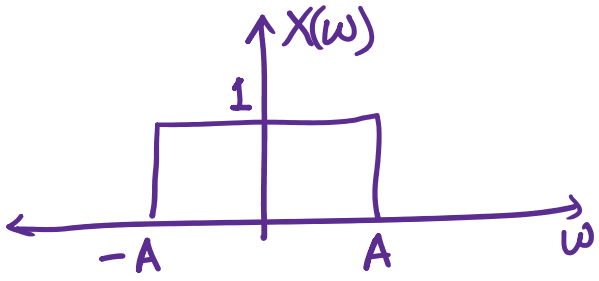
$\longleftrightarrow X(\omega)?$



$$X(\omega) = \int_{-T_1}^{T_1} 1\cdot e^{-j\omega t}\,dt = \frac{e^{-j\omega T_1} - e^{+j\omega T_1}}{-j\omega} = \frac{2\sin(\omega T_1)}{\omega}$$

$$=$$

$\longleftrightarrow$



$$x(t) = \frac{1}{2\pi}\int_{-A}^{A} 1\cdot e^{j\omega t}\,d\omega = \frac{e^{jtA} - e^{-jAt}}{2\pi jt} = \frac{\sin(At)}{\pi t}$$

Properties:

1) Linearity & Invertible

$$x(t) \longrightarrow \boxed{\text{F.T.}} \longrightarrow X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt$$

2) $x(t)$ is real $\Longrightarrow X(-\omega) = X^*(\omega)$

Pf: $X(-\omega) = \int_{-\infty}^{\infty} x(t) e^{j\omega t}\, dt = \int_{-\infty}^{\infty} x^*(t) e^{j\omega t}\, dt = X^*(\omega)$

$\Rightarrow$ $|X(\omega)|$ is symmetric

& $\angle X(\omega)$ is antisymmetric

3) $x(t)$ is real & even $\Longrightarrow X(\omega)$ is real & even

Pf: $\left.\begin{array}{l} x(-t) \longleftrightarrow X(-\omega) \\ \therefore x(t) \longleftrightarrow X(-\omega) = X^*(\omega) = X(\omega) \end{array}\right\} \Rightarrow X(\omega)$ is real & even